Ratios and percents tutor evidence map

My role AI tutor math evaluation

I mapped productive and problematic reasoning about ratios and percents using 31 anchor questions and 95 eligible tutor conversations. The evidence sets boundaries for what the tutor can infer.

ORIGINAL WORK FILE EXCERPT

AUTONOMOUS CONVERSATION-INFORMED FACET MAP

Map 26: Ratios, Percents, and Proportional Reasoning

Grades 7–8 | Autonomous, conversation-informed facet map

Standards: 7.MA.NNR.2 • 8.MA.NNR.2

Working-final autonomous set: 5 Map-26-owned productive facets and 7 problematic facets. Equivalent ratio/rate scaling, percent-of-quantity, inverse part–whole reasoning, and multistep operation sequencing reuse already validated facets from Maps 24, 25, and 30. This keeps the map focused on small beliefs: what the two ratio terms refer to, how a part–part ratio relates to a whole, how a combined total is partitioned by ratio units, which quantity is the reference whole for a percent, and why percent change is relative to the original amount.

Scope and evidence base

7.MA.NNR.2: Recognize the form of a number appropriate for use in a given situation and apply ratio, percent, and proportion in a variety of situations.

8.MA.NNR.2: Recognize the form of a number appropriate for use in a given situation and apply ratio, percent, and proportion in a variety of situations.

Use of generated questions: Generated questions were used to discover recurring student reasoning and refine evidence boundaries.